

Kaggle TensorFlow Speech Recognition Challenge

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Abstract

Keyword spotting (KWS) is a core component of always-on speech interfaces, where both robustness and low-latency inference are required. This paper presents a controlled comparative study of KWS architectures on the Kaggle TensorFlow Speech Recognition Challenge corpus under a unified training and evaluation protocol. We formulate a fixed 12-class task (10 commands plus `__unknown__` and `__silence__`) and investigate 3 factors that govern performance: front-end representation, architectural inductive bias, and non-command modeling strategy. To isolate architectural effects, all systems are trained from random initialization with identical optimization controls, deterministic settings, and multi-seed evaluation. The experimental design follows a 4-phase selection procedure: feature ablation, global optimization tuning, backbone comparison across model families, and targeted non-command methodology analysis. Final held-out testing is restricted to systems selected in the last phase, including both single-model and within-method ensemble variants. We report the executed protocol and resulting comparative outcomes under this controlled setup.

Index Terms: keyword spotting, Kaggle TensorFlow Speech Recognition Challenge, MLOps, convolutional neural networks, transformers, state-space models, recurrent models

1. Introduction and Motivation

Detecting a fixed, limited vocabulary within an audio stream occupies a unique and highly resource-constrained niche in modern speech processing [1, 2]. Recent progress has diversified the model landscape, but many comparisons still confound architectural differences with inconsistent preprocessing and optimization choices. Our objective is to evaluate architectures under a common, from-scratch training protocol, ensuring that observed differences are attributable to structural inductive biases rather than transfer-learning advantages [3].

Our study is structured around three primary research questions, designed to systematically isolate the drivers of KWS efficacy:

- **Q1 (Feature strategy):** Which feature extraction pipeline produces the most generalizable representations for a fixed 1-second classification task? This evaluates whether the default reliance on standard log-mel spectrograms is genuinely optimal.
- **Q2 (Architectural inductive bias):** Which structural prior best captures the temporal-spectral dynamics of brief KWS inputs? We directly compare local spatial hierarchies, global all-to-all attention, and finite-memory sequential compression.
- **Q3 (Non-command class methodology):** Which methodol-

ogy best recognizes `__unknown__` and `__silence__` while preserving macro-F1 on the core command classes?

To prevent hyperparameter explosion and to contextualize model evaluation within a practical engineering framework, we structure our experiments into a strict 4-phase pipeline (introduced in Section 3). In this manuscript version, the emphasis is methodological: the Results section specifies the held-out test protocol, while empirical comparisons are reported after all runs complete.

2. Dataset and Task Formulation

To provide a direct evaluation of raw architectural capacity, this project leverages the Kaggle TensorFlow Speech Recognition Challenge dataset (derived from Google Speech Commands v1) [4] as a controlled benchmark. These constraints remove substantial domain-level complexities that typically confound architectural comparisons.

We use a fixed 12-class taxonomy: the target commands are {yes, no, up, down, left, right, on, off, stop, go}, `__silence__` is mapped from `background_noise_`, and every other spoken label is merged into `__unknown__`. Dataset preprocessing is standardized as follows: (i) long background-noise recordings are split into one-second silence samples, (ii) short utterances are zero-padded to one second, (iii) samples longer than one second are excluded in strict one-second dataset modes, and (iv) waveform loading enforces a fixed one-second input window for every sample (with padding or truncation as required). Consequently, all downstream feature extraction and modeling stages operate on a uniform temporal support.

Split construction follows three policies. First, command-only small splits are built with fixed per-class quotas (1,000 train, 250 validation, 250 test for each of the 10 commands), and are used in Phases 1–3. Second, extended splits for Phase 4 include all 12 classes and are balanced at the command-vs-unknown level per split. Third, unknown examples used in extended splits are deduplicated by audio-content hash and additionally capped to at most 20,000 samples per split to avoid unknown-class dominance.

Table 1 reports exact split sizes after applying this 12-class remapping to the repository split lists. We provide total sample count, active class count, command/non-command composition, and the intended usage of each split in the experimental pipeline.

2.1. Unknown-Class Synthesis Methodology

Before merging labels, we persist a metadata artifact `train/split_lists/unknown.origin.labels.csv` that maps each merged unknown file to its original source

Table 1: *Dataset split statistics under the 12-class mapping (10 commands + unknown + silence).*

Split	Total N	Active classes	Command samples	Unknown samples	Silence samples	Used for
train.small	10,000	10	10,000	0	0	Phases 1–3 train
valid.small	2,500	10	2,500	0	0	Phases 1–3 validation
test.small	2,500	10	2,500	0	0	Phases 1–3 diagnostics
train.extended	38,210	12	18,946	18,946	318	Phase 4 train
valid.extended	4,776	12	2,368	2,368	40	Phase 4 validation
test.extended	4,776	12	2,368	2,368	40	Phase 4 test

label, ensuring no provenance is lost. For the extended dataset configuration, the unknown pool is built from the original labels present under `train/audio` excluding `_background_noise_`, `_unknown_`, and the padded directory. In practice, this means both command and non-command source clips can contribute to synthetic `_unknown_` mixtures, provided the clips in one mixture come from different classes.

Each synthetic sample is formed by blending 2 or 3 one-second clips. Before mixing, each source is mean-centered, RMS-normalized toward a shared target level, and lightly smoothed with a short delayed component to reduce abrupt local transients. The final mixture is then produced by dense short-frame overlap where all sources remain active throughout the full second with slowly varying weights, encouraging spectro-temporal continuity rather than hard segment boundaries.

To avoid regenerating the same example, uniqueness is tracked at the source-file level (sorted source-path tuple), not at the class-combination level. After synthesis, each candidate is passed through an envelope-based smoothness gate (windowed RMS trajectory checks for abrupt jumps and deep internal valleys). Candidates that fail this gate are discarded and regenerated in a retry loop; only accepted candidates are written to the `_unknown_` directory. During extended-split construction, original unknown candidates are first deduplicated by audio-content hash; if this deduplicated pool is smaller than the required unknown quota, synthetic mixtures are added as a top-up until the target unknown count is reached (or generation attempts are exhausted).

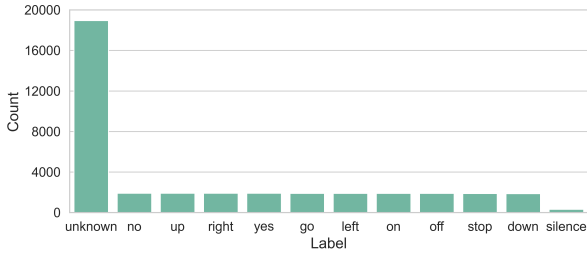


Figure 1: *Class distribution in the extended training set.*

2.2. Audio Analysis

In analyzing the raw audio data, we focus on understanding the raw waveforms and their spectral properties. Figure 2 shows an example of a raw audio waveform and its corresponding spectrogram, illustrating the temporal and frequency characteristics of a typical command sample. The spectrogram reveals the energy distribution across frequencies over time, which is crucial for feature extraction strategies. Raw waveforms exhibit high variability in amplitude and temporal structure, while spectro-

grams provide a more stable representation that captures the essential characteristics of the audio signal. Observing raw waveforms of different classes (Appendix A) reveals that amplitude patterns vary; however, the main signal typically begins around 0.4 seconds.

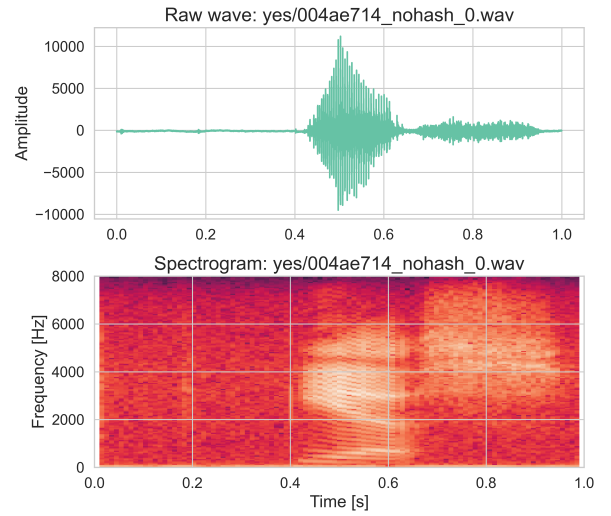


Figure 2: *Raw wave and spectrogram example from train set for "yes".*

Another factor to consider is the average spectral content across different command classes. Figure 3 shows the mean FFT magnitude for each label in the training set, highlighting differences in frequency content between commands. This suggests that commands have characteristic spectral profiles that models can exploit for discrimination.

3. Experimental Design and MLOps Pipeline

To systematically address our research questions without combinatorial explosion, we filter configurations through a gated, 4-phase funnel. Each phase selects the best setup based on macro-F1 across 3 fixed random seeds, and this configuration is carried forward to the next stage. Table 2 outlines the experiment scope and compute budget. Split usage is fixed by the dataset protocol in Section 2: Phases 1–3 use `small` splits, while Phase 4 uses `extended` splits.

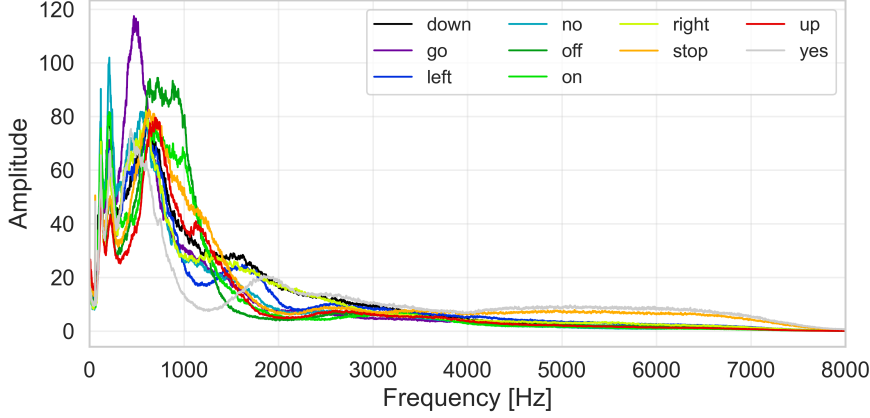


Figure 3: Mean FFT magnitude per label in the training set.

Table 2: Executed trial scope per experimental phase (one trial may train multiple internal components).

Phase	Experiment Scope	Configs	Seeds	Total Runs
1	Feature Strategy Ablation (5 features $\times$ 2 proxy models)	10	3	30
2	Global Hyperparameter Tuning (best feature $\times$ 2 proxies $\times$ 4 optim. combos)	8	3	24
3	Inductive Bias Comparison (AST: 8, ConvNeXt: 2, MLP-Mixer: 4; planned SSAMBA and xLSTM excluded)	14	3	42
4	Non-Command Method Comparison (5 strategies $\times$ 3 top models on __unknown__/__silence__)	15	3	45
Total	Cumulative Trial Executions	47		141

3.1. Uniform Training Protocol

To ensure fair architectural comparison, all models are trained under an identical, deterministic protocol with shared hyperparameters and optimization controls. Model training follows AdamW ([5]) with fixed base settings ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$) and a base learning rate of 3×10^{-4} . Phase 2 searches the AdamW regularization and scheduler settings, and the winning optimizer configuration is then carried forward into Phases 3 and 4. Gradient clipping is applied with a maximum norm threshold of 1.0 to stabilize training on recurrent and attention-based architectures.

The training regimen uses a batch size of 32 (a constraint forced by hardware limitations) across all experiments and model families, with unlimited training epochs and a learning-rate scheduler employing cosine annealing with linear warmup (cosine-annealing-warmup). A warmup of 5 epochs lin-

early increases the learning rate from a near-zero start factor of 10^{-3} , followed by cosine decay to a minimum rate of 10^{-6} . Early stopping is enforced with a patience of 10 epochs without macro-F1 improvement on the validation set. Deterministic execution is enabled via fixed seed initialization, and validation is performed every epoch to track model convergence.

3.2. Phase 1: Feature Extraction Strategy (Q1)

Raw audio waveforms are highly inefficient for direct neural network processing due to a lack of semantic meaning at the sample level and high adjacent correlation [6]. We transform the signal into time-frequency representations using the Short-Time Fourier Transform (STFT) [7].

Feature extraction is governed by the time-frequency uncertainty principle (Gabor limit), which bounds resolution such that $\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$. We encapsulate all preprocessing pipelines within PyTorch `nn.Modules` to ensure differentiable, GPU-accelerated computation, evaluating five distinct strategies:

Mel-Spectrogram (Baseline): Emulating human auditory non-linearity via the Mel scale ($m = 2595 \log_{10}(1 + f/700)$) [8], this baseline uses $n_{fft} = 1024$ (64 ms window) and a 10 ms stride. Power spectra are filtered and compressed via an 80 dB-bounded log-amplitude transform to stabilize gradients.

High Temporal Resolution Mel: Shifting the Gabor limit temporally, this setup halves the window and hop sizes (32 ms window; 5 ms stride). The 200 Hz frame rate exposes sub-phonemic transients otherwise smeared by wider windows.

MFCC: Applying a Discrete Cosine Transform (DCT) to log-mel energies projects the signal into the quefrequency domain. Retaining 40 coefficients decorrelates channels [9] and slashes the data footprint by $\sim 93\%$.

PCEN: Per-Channel Energy Normalization [10] applies dynamic automatic gain control. It suppresses stationary noise and emphasizes transients [11], enhancing robustness against varied SNRs.

Mel + SpecAugment: Acting as structured input dropout, SpecAugment [12] applies stochastic rectangular masks across time and frequency axes, forcing the model to learn distributed phonetic representations.

To prevent favoring a specific architecture during this phase, we

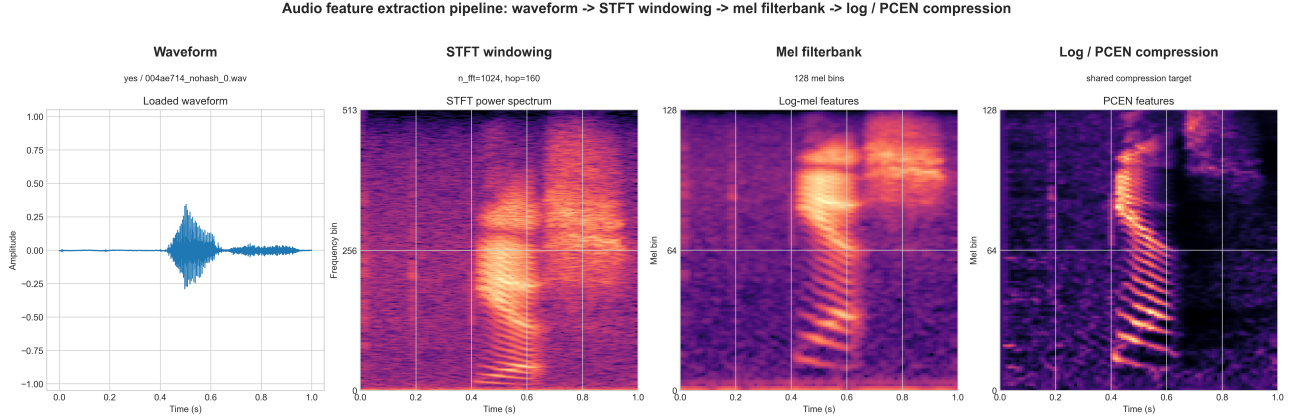


Figure 4: Audio feature extraction pipeline from waveform loading through STFT windowing, mel filterbank projection, and log/PCEN compression.

evaluate the pipelines on two proxy baselines: a spatial ConvNeXt model and a sequential xLSTM model. ConvNeXt is implemented with a 1-channel input stem and shared adapter-level shape normalization, which keeps all listed feature variants (including 40×101 MFCC) executable in the current stack. This ensures we do not inadvertently handicap sequential models that might uniquely benefit from high-temporal strides. Table 3 and Figure 4 detail these setups.

3.3. Phases 2 & 3: Architectural Bias and Optimization (Q2)

To isolate structural effects, we train all models entirely from random initialization. Rather than asking which pre-training source transfers best, this design evaluates which architecture learns most effectively from the supervision signal itself.

Phase 2 tunes the global AdamW configuration (Table 4) on the best Phase 1 feature set. Phase 3 then sweeps model-specific hyperparameters (Table 5) using the winning Phase 2 optimizer settings across the executed families. The original plan also included xLSTM as a high-capacity recurrent comparator, but it was excluded from Phase 3 execution after Phases 1 and 2: relative to ConvNeXt, xLSTM showed weaker proxy-stage behavior while requiring more than $10\times$ longer training time. SSAMBA was likewise excluded from executed Phase 3 runs after an initial round on the target hardware: the available macOS Mamba backend lacks a fused parallel scan, and observed wall-clock cost exceeded 3 hours per epoch, which was incompatible with the multi-trial, multi-seed Phase 3 budget.

AST (Audio Spectrogram Transformer): Implemented with Hugging Face `ASTForAudioClassification` (from `transformers`) using `ASTConfig` with hidden size 512, 8 encoder layers, 8 attention heads, and intermediate size 2048 (targeted ~ 25 M-scale setup in this work).

Key feature / why chosen: global self-attention from the first layer gives immediate all-to-all context, making AST the explicit “global dependency” reference model [13, 3].

ConvNeXt (Modern CNN): Implemented as `torchvision.models.convnext_tiny` (primary backend), with the first stem convolution adapted to one-channel spectrogram input; `timm convnext_tiny` is kept as backend fallback.

Key feature / why chosen: strong locality and translation-

equivariant inductive bias provide the canonical modern CNN baseline against which global and recurrent models are compared [14, 15].

SSAMBA (Selective State Space; planned, excluded):

Implemented through `mamba_ssm.Mamba` blocks in a repository wrapper (`MambaHead`), stacked with configuration $d_{\text{model}} = 768$, $d_{\text{state}} = 64$, $d_{\text{conv}} = 4$, $\text{expand} = 2$, and 6 layers (targeted ~ 25 M-scale setup in this work). *Key feature / why chosen:* input-selective state-space updates target long-range sequence modeling with linear-time scaling, representing an efficient alternative to full attention [16, 17].

This branch was excluded from Phase 3 execution because it required more than 3 hours per training epoch.

xLSTM (High-Capacity Recurrence; planned, not executed):

Implemented with `xlstm.xLSTMBlockStack` in a repository wrapper (`xLSTMHead`), using embedding dimension 704/768 and 8 blocks (alternating mLSTM/sLSTM placements; targeted ~ 25 M-scale setup).

Key feature / why chosen: matrix-memory recurrence $C_t \in \mathbb{R}^{d \times d}$ increases recurrent capacity beyond standard LSTM cells, serving as the originally planned high-capacity recurrent comparator [18].

This branch was excluded from Phase 3 execution because its Phase 1–2 proxy behavior was worse than ConvNeXt while its training time was more than $10\times$ longer.

MLP-Mixer: Implemented as `timm gmixer_24_224` (official gMLP/Mixer-family model) for the MLP-only baseline.

Key feature / why chosen: token/channel mixing via dense projections with no explicit convolutional or recurrent prior acts as an architectural “minimal prior” control [19].

The counts in Table 5 are measured on the repository’s adapter-ized backbones, not on the upstream libraries’ unmodified reference models. This keeps the comparison honest about the actual code we run while still keeping the backbones in a comparable parameter band; the point is adapter-level parity, not matching an external giant-model budget.

3.4. Phase 4: Non-Command Class Methodology (Q3)

All core models are optimized natively via standard `CrossEntropyLoss` [20]. We intentionally defer imbalance-specific loss shaping and resampling in earlier

Table 3: Feature extraction strategies compared in Phase 1.

Strategy	n_fft	hop_length	Window	Hop	Frames	Bins	Approx. input size (1 s)	Augmentation
Mel (baseline)	1024	160	64 ms	10 ms	101	128 mels	12,928 (128 × 101)	none
Mel (hi-temporal)	512	80	32 ms	5 ms	201	128 mels	25,728 (128 × 201)	none
MFCC	1024	160	64 ms	10 ms	101	40 MFCCs	4,040 (40 × 101)	none
PCEN	1024	160	64 ms	10 ms	101	128 mels	12,928 (128 × 101)	none
Mel + SpecAugment	1024	160	64 ms	10 ms	101	128 mels	12,928 (128 × 101)	freq+time masking

Table 4: Global optimization hyperparameters (Phase 2).

Parameter	Values	Purpose
AdamW learning rate	3×10^{-4}	Fixed base step size reused by later phases
AdamW β_1, β_2	(0.9, 0.999)	Fixed first- and second-moment coefficients
AdamW ϵ	10^{-8}	Fixed numerical stability constant
Weight decay	$10^{-2}, 10^{-1}$	Swept AdamW regularization strength
LR scheduler	Cosine Annealing with Warmup / Reduce LR on Plateau	Swept scheduler family; selected schedule is reused later

phases to keep architectural comparisons interpretable. Phases 1–3 use command-only data splits while preserving a shared implementation template for model heads. Phase 4 keeps the selected backbone family and hyperparameters fixed and retrains on `train_extended` for explicit non-command handling. In this sense, “freezing the top 3 architectures” refers to freezing architecture and optimization choices selected upstream. In Phase 4, we test 5 alternatives for handling `__unknown__` and `__silence__`:

1. **Flat multiclass baseline:** single-head 12-class classification.
2. **Sampling control:** weighted random sampling with replacement and fixed target class priors: $p_{\text{unknown}} = 0.25$, $p_{\text{silence}} = 0.15$, and $p_{\text{cmd},k} = 0.06$ for each of the 10 command labels (summing to 1.0).
3. **Loss reweighting:** weighted cross-entropy with elevated penalties on non-command misclassification.
4. **Two-stage detector:** a separate binary gate (command vs. non-command), with command and non-command branch probabilities fused through gate-conditioned composition.
5. **Shared two-head model:** a command head (10 labels) plus a dedicated non-command head (silence vs. unknown).

For sampling control, if class c has n_c training examples, each example from that class receives weight

$$w_c = \frac{p_c}{n_c}.$$

Each epoch draws N samples (with replacement), where N equals the original training-split size. In the implemented Phase 4 loader, inverse-frequency weighting with explicit silence up-weighting is applied as a base balancing mechanism, and strategy-specific controls are layered on top (e.g., strategy 2 priors and strategy 3 loss weights). Unknown-sample synthesis remains conditionally enabled: additional synthetic unknown samples are generated only when empirical unknown support falls below the mean support of the command classes.

To keep comparisons fair, all methods reuse identical data splits and 3 fixed seeds (0, 42, and 2003). Model selection follows a strict gate: any method that degrades core-command macro-F1 by more than $\delta = 1.0$ percentage point relative to its corresponding Phase 3 command baseline is rejected. Among remaining candidates, we maximize non-command macro-F1,

$$\text{Macro-F1}_{\text{NC}} = \frac{\text{F1}_{\text{unknown}} + \text{F1}_{\text{silence}}}{2},$$

and use silence false-trigger rate and unknown-to-command leakage as tie-breakers.

The outputs of Phase 4 define the final deployable systems and are the only systems advanced to held-out test-set validation. Strategy 4 is executed as one trial configuration that internally trains separate gate, command, and non-command components; Table 2 reports trial-level counts rather than internal component fits. We evaluate 3 inference flows: (i) single-stage multiclass flow (input $\rightarrow$ one model $\rightarrow$ class), (ii) two-stage detector flow (input $\rightarrow$ binary gate and branch experts $\rightarrow$ gate-weighted probability fusion), and (iii) shared-backbone two-head flow (input $\rightarrow$ shared encoder $\rightarrow$ command head and non-command head, with a fixed decision rule).

4. Executed Scope, Reproducibility, and Training Control

The original experimental plan defined a four-phase funnel: feature strategy selection, global optimizer tuning, architecture comparison, and final non-command handling. The executed scope keeps this structure intact for Phases 1, 2, and 4, while Phase 3 is narrowed to AST, ConvNeXt, and MLP-Mixer. The xLSTM branch remains part of the planned design and repository implementation, but it is explicitly excluded from the executed Phase 3 sweep because its Phase 1–2 proxy behavior was worse than ConvNeXt while training took more than $10\times$ longer. SSAMBA is similarly excluded from execution: an initial Phase 3 round on the target hardware measured wall-clock cost above 3 hours per epoch (the available macOS Mamba backend has no fused parallel scan), which would have been incompatible with the planned multi-seed budget; the architecture, sweep dimensions, and adapter implementation are nevertheless retained for future runs on CUDA hardware. Consequently, the executed Phase 3 grid contains 14 configurations (42 seeded runs), rather than the originally planned 30 configurations (90 seeded runs).

To account for hardware-level non-determinism during GPU acceleration, we execute every experiment three separate times using fixed starting points (seeds 0, 42, and 2003) and report aggregated results (mean $\pm$ standard deviation).

All experiments are run on a single Apple MacBook Pro with an M4 Pro chip, 48 GB unified memory, and a 1024 GB SSD. This hardware budget motivates the use of `train_small` and `valid_small` in Phases 1–3. Figure 5 summarizes the wall-time breakdown and run counts per phase;

Table 5: *Model-specific hyperparameter sweeps (Phase 3), including architecture type and parameter count from the repo-specific adapters used in this work (default non-pretrained configuration, 12 output classes). SSAMBA and xLSTM are retained as planned scope but excluded from executed runs.*

Model	Arch. type	Total params	Sweep Space
AST	Transformer (global attention)	25,416,204	Dropout {0.1, 0.5}; Head {linear, MLP(256)}; Pos. Emb. {interp., learned}; fixed size: hidden = 512, layers = 8, heads = 8, intermediate = 2048 [3]
ConvNeXt	CNN (local spatial hierarchy)	27,826,284	Stoch. Depth {0.0, 0.2}; Kernel size $\{7 \times 7\}$ [15]
SSAMBA	State-space sequence model	23,963,148	Planned only; excluded from executed Phase 3 runs after an initial round showed wall-clock cost above 3 hours per epoch on the target hardware (macOS Mamba backend lacks a fused parallel scan), incompatible with the multi-seed Phase 3 budget. Original sweep: Pooling {mean, max}; CLS {yes, no}; Stride {10ms, 5ms}; fixed size: $d_{\text{model}} = 768$, $d_{\text{state}} = 64$, expand = 2, layers = 6 [16, 17]
xLSTM	Recurrent sequence model	24,259,180	Planned only; excluded from executed Phase 3 runs after weaker Phase 1–2 proxy behavior than ConvNeXt and more than $10\times$ longer training time. Original sweep: Dim $d \in \{704, 768\}$; State Reset {T, F}; Output {final, mean}; fixed depth: blocks = 8 [18]
MLP-Mixer	MLP-only token/channel mixer	24,144,108	Dropout {0.0, 0.2}; Head L2 Norm {yes, no}; default timm variant: gmixer_24_224 [19]

the total compute time accumulated to around 320 GPU-hours across all phases. Table 6 breaks the same tracking data down by model architecture and reports the average step speed observed for each family.

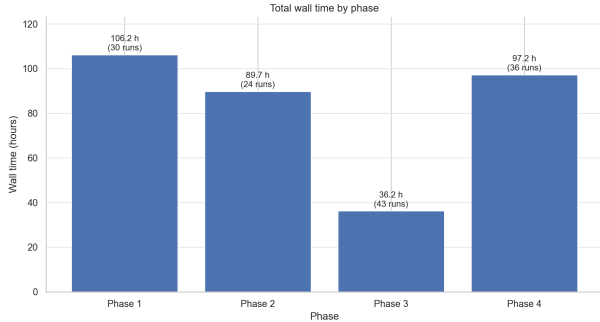


Figure 5: *Total wall time and run count by phase, measured from the MLflow tracking database.*

To stabilize compute and comparisons in Phases 1–3, the small splits are fixed to exactly 1,000/250/250 samples per command class for train/validation/test, respectively.

We employ AdamW [5] to explicitly decouple weight decay from adaptive gradient scaling, and rely on Kaiming normal initialization [21]. To guarantee that experiments can be perfectly duplicated, we enforce deterministic data loading and automatically generate a `ReproducibilityReport` via MLflow [22] for every run. This report logs all environmental conditions, library versions, and the precise state of the codebase.

5. Results

Final test reporting covers all completed Phase 4 trials, since this stage determines deployment-relevant behavior for `__unknown__` and `__silence__`. After the Phase 4 sweep completes, each trial is evaluated once on the held-out test set. We

report per-class precision, recall, and F1, together with command macro-F1 and non-command macro-F1.

Beyond single-model results, we evaluate all ensemble combinations across the 3 Phase 3-derived backbones within each fixed Phase 4 method. Denoting model outputs by M_1, M_2, M_3 , we test all non-empty subsets (i.e., $2^3 - 1 = 7$ combinations) and aggregate class probabilities by simple averaging under a common within-method calibration rule. Cross-method ensembles are not considered; each ensemble is built from models produced by a single Phase 4 strategy. These ensemble checks are inference-only and therefore excluded from the cumulative training execution counts in Table 2.

All results are available on Google Drive.

5.1. Phase 1

Phase 1 focuses on evaluating different feature extraction strategies. Results are reported in terms of macro-F1 for each strategy across the two proxy models (ConvNeXt and xLSTM), and are visualized in Figure 6. Mel-spectrogram and MFCC variants outperform the other feature extraction methods, although satisfactory performance is observed only on ConvNeXt. xLSTM shows a significant performance drop across all feature extraction strategies, which motivates its exclusion from later phases due to both lower competitiveness and higher computational cost.

5.2. Phase 2

The main focus in this section is global hyperparameter tuning. The best-performing optimizer configuration is selected based on macro-F1 and convergence speed. Figure 7 compares convergence speed across optimizer configurations. Convergence differences are small, with Reduce LR on Plateau converging slightly faster than cosine annealing with warmup, but not enough to establish a decisive winner on speed alone. Weight decay 10^{-2} provides slightly better mean macro-F1 than 10^{-1} , and 10^{-1} shows higher standard deviation on ConvNeXt, indicating reduced stability.

Table 6: Average model step speed by architecture. Transformer- and CNN-based models show superior step speed compared to recurrent and MLP-only models.

Architecture	Runs	Mean step time (ms)	Median step time (ms)	Mean steps/s	Total steps
AST	36	6.26	4.86	210.17	823,484
ConvNeXt	45	5.55	3.68	262.67	1,011,138
MLP-Mixer	24	26.84	27.06	49.74	300,696
SSAMBA	1	13,199.47	13,199.47	0.08	700
xLSTM	27	56.62	49.75	20.94	333,971

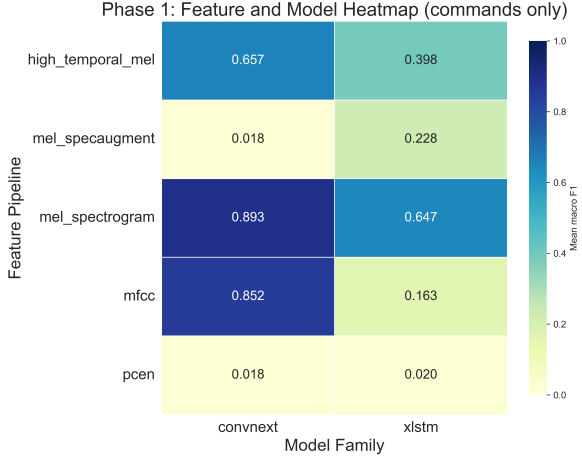


Figure 6: Phase 1 results: macro-F1 scores for each feature extraction strategy across the two models.

The xLSTM results are interpreted as secondary evidence because performance is substantially worse than ConvNeXt, and all tested combinations are closely clustered. Overall F1 is similar across xLSTM regularization settings (Figure 18).

5.3. Phase 3

With the best training setup selected in Phases 1 and 2, we evaluate three models in Phase 3: AST, ConvNeXt, and MLP-Mixer. Results are reported as macro-F1 across the tested hyperparameter configurations (Figure 8). ConvNeXt shows the strongest and most stable performance. AST reaches comparable performance in selected configurations but with higher variance. MLP-Mixer performs worst across configurations, with a substantial macro-F1 drop relative to AST and ConvNeXt. Failure highlights in Figure 9 show how often models fall below random guessing on specific classes: ConvNeXt has zero such cases, AST has two, and MLP-Mixer exceeds 50%.

In terms of efficiency between ConvNeXt and AST, ConvNeXt’s total training time is significantly lower than AST’s, which, paired with its better performance, makes it the strongest choice for this task (Figure 20). Analyzing training loss versus epoch (Figure 10) shows that ConvNeXt exhibits a more stable and consistent loss decrease than AST and MLP-Mixer, which show larger fluctuations and higher final losses.

Hyperparameter sensitivity analysis for AST (Figure 11) shows consistent mean performance for the linear head across both dropout values and positional embedding strategies. For MLP-Mixer, dropout 0.5 leads to a significant performance drop compared to dropout 0.1, which may indicate instability

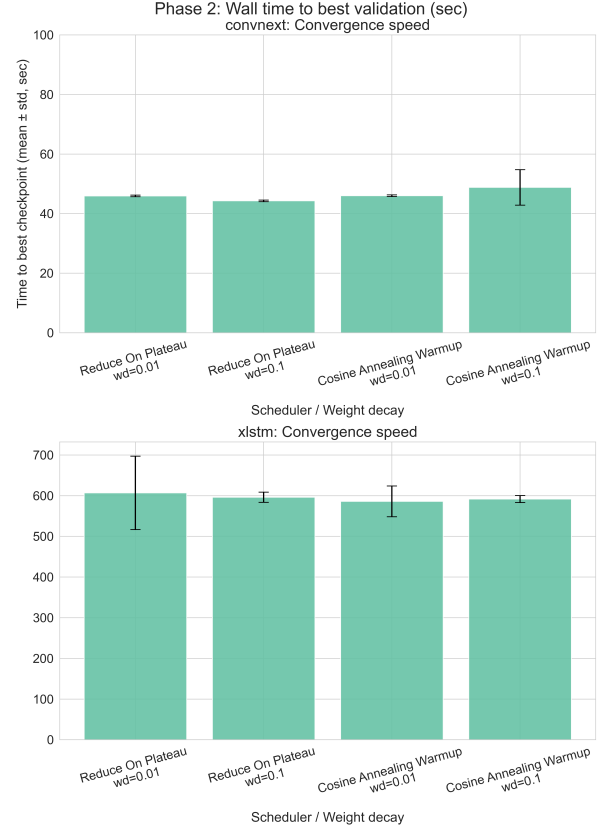


Figure 7: Phase 2 results: convergence speed of different optimizer configurations.

in this model family.

5.4. Phase 4

Phase 4 focuses on evaluating strategies for handling non-command classes (`_unknown_` and `_silence_`). Besides macro-F1 on the non-command classes, we evaluate silence false-trigger rate and unknown-to-command leakage (how often unknown samples are misclassified as command labels). The best-performing approach is *flat multiclass*, with *loss reweighting* showing similar performance (Figure 12). *Sampling control* on ConvNeXt also achieves good results, but drops substantially on AST. The worst-performing strategy is *two-stage detector*, which shows a clear command macro-F1 drop and a very high silence false-trigger rate. Figure 13 confirms this failure mode, while flat multiclass and sampling control maintain lower false-trigger rates and better robustness against silence-to-command

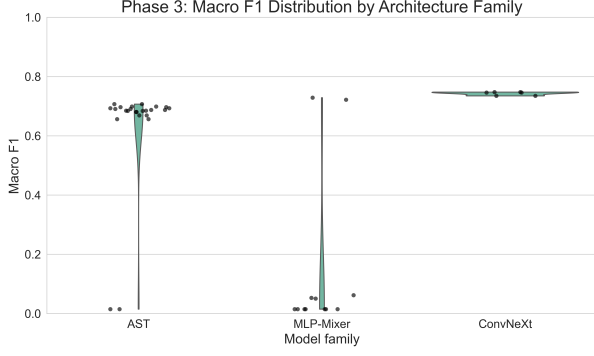


Figure 8: *Phase 3 results: architecture distribution of the best performing models.*

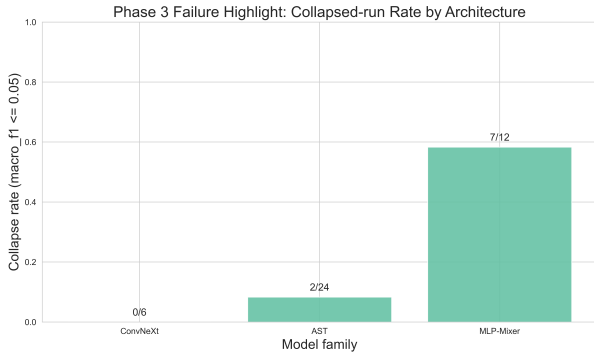


Figure 9: *Phase 3 results: failure cases highlighted for the best performing models.*

errors. MLP-Mixer again proves to be a weak choice for this task. Additional score and false-trigger comparisons are shown in Figure 19.

5.5. Class Comparison

To further analyze model behavior, we compare per-class precision, recall, and F1 for the best-performing model (ConvNeXt with flat multiclass strategy) across command and non-command classes (Figure 14). The best model achieves F1 above 0.85 for all classes, with macro-F1 of 0.917. The most difficult labels are “no” and “go”. The “unknown” class performs strongly, with one of the highest F1 scores among all classes.

6. Conclusion

This study delivers a complete, reproducible 4-phase benchmark for keyword spotting under a fixed 12-class formulation (10 commands + `__unknown__` + `__silence__`). Across the full pipeline, ConvNeXt emerged as the strongest and most stable architecture, with AST as a competitive but less stable alternative and MLP-Mixer consistently underperforming. In feature selection, mel-based front-ends (including MFCC as a strong compact variant) clearly outperformed the remaining candidates. In global optimization tuning, cosine annealing with warmup combined with weight decay 10^{-2} provided the best convergence-quality trade-off for the winning branch.

For non-command handling, the flat multiclass strategy pro-

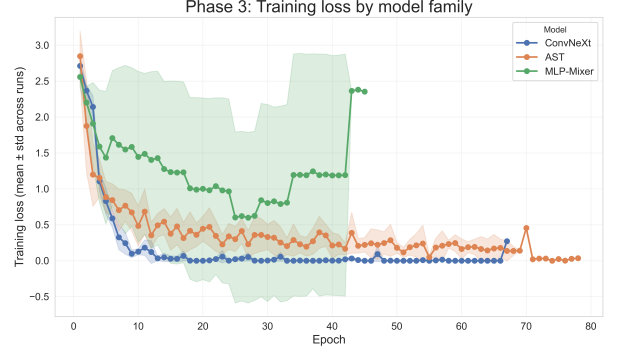


Figure 10: *Phase 3 results: Training loss vs. epoch for different model families.*

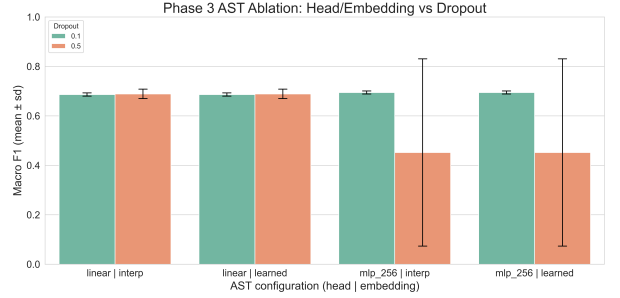


Figure 11: *Phase 3 results: Ablation study for AST.*

vided the best overall deployment profile, while loss reweighting remained a close alternative. In contrast, the two-stage detector showed a clear failure mode through elevated silence false-trigger behavior, making it unsuitable under the strict command-preservation gate used in this work. Per-class diagnostics further showed that command pairs with higher phonetic similarity (notably “go/no” and “on/off”) remain the main source of residual error, even in the best model.

Methodologically, the main contribution is not only the final ranking but the controlled decision funnel itself: each phase narrows the search space while preserving comparability through fixed seeds, shared optimization controls, and explicit non-command risk metrics (unknown leakage and silence false triggers). Although xLSTM and SSAMBA remain implemented as planned branches, their exclusion from executed Phase 3 runs is justified by the measured compute-performance trade-off on the target hardware. Overall, the presented protocol and results provide a practical template for building robust KWS systems under realistic resource constraints.

7. Future Work

While this study establishes a strictly controlled baseline for evaluating KWS architectures, several avenues remain for future exploration. First, because our methodology isolates architectural inductive biases by training entirely from random initialization, future work could compare these baselines against pre-trained variants to quantify the performance gains afforded by transfer learning. Second, although our data preprocessing enforces a uniform one-second temporal support, adapting the top-performing architectures to process variable-length inputs or continuous, untrimmed audio streams would better simulate

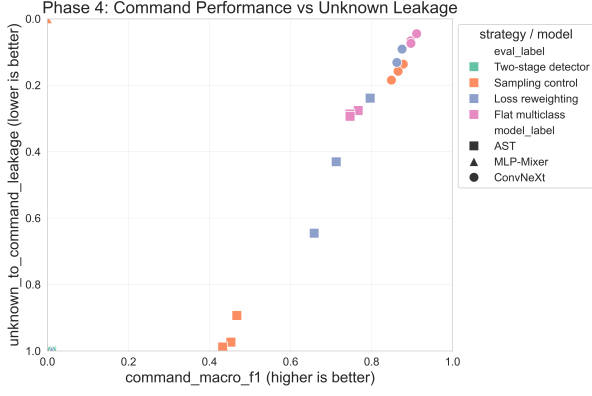


Figure 12: Phase 4 results: Command scores vs. Non-command scores for different strategies.



Figure 13: Phase 4 results: Silence false trigger rates for different strategies.

true always-on deployment conditions. Third, while the synthesis of the `__unknown__` class is currently limited to fixed extended splits, deploying our proposed synthesis pipeline as a dynamic, on-the-fly augmentation mechanism could further harden models against out-of-vocabulary spoken utterances. Finally, our current ensemble evaluation is restricted to within-method combinations; future research could explore cross-method ensembles to determine if heterogeneous non-command handling strategies yield superior robustness against false triggers. Due to time constraints under the available hardware budget, we did not deliver a full cross-method ensemble study in the final reporting cycle; this remains an explicit follow-up item.

8. AI Disclosure

This report was edited with AI assistance, to improve clarity, consistency, and formatting. Part of the docstrings, some phrasing and code refactoring or debugging were refined using AI tools, but all scientific decisions and interpretations are the authors' own. The core content, experimental design, and analysis were developed by the authors.

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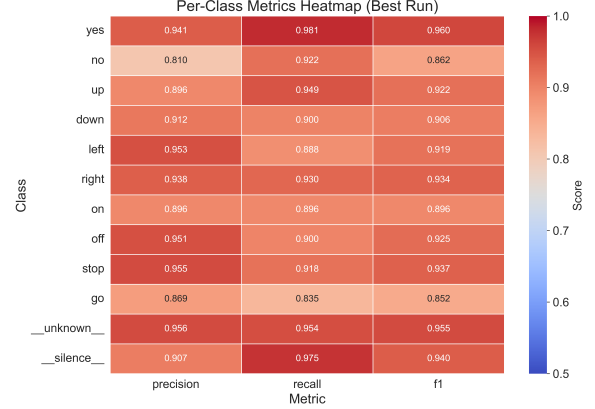


Figure 14: Phase 4 results: Per-class performance for the best performing model.

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A. Additional Analyses and Visualizations

Figures 15, 16, and 17 provide additional waveform–spectrogram examples for the labels *down*, *no*, and *up*, respectively. These supplementary plots complement the main text by illustrating cross-class variability in temporal envelope shape, onset location, and spectral energy concentration. Together, they support the motivation for using time-frequency features and for comparing architectures with different inductive biases for local versus global temporal-spectral structure.

Figures 18 and 19 provide additional visualizations of the regularization sensitivity across different configurations and the score distribution across different strategies, respectively. Figure 20 shows the training efficiency for different strategies, which is an important factor to consider when selecting a model for deployment.

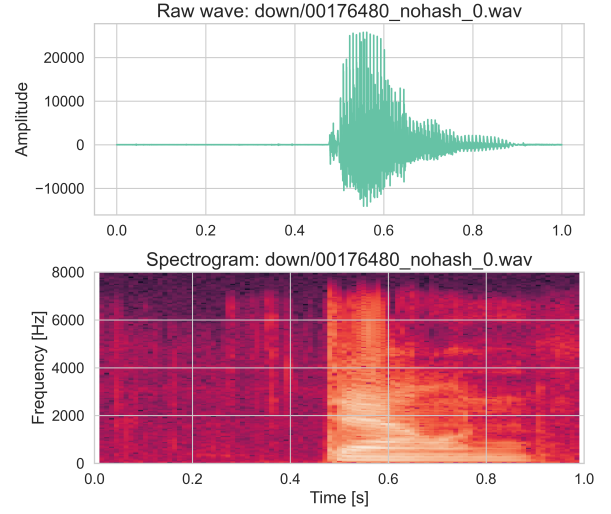


Figure 15: Raw wave and spectrogram example from train set for “down”.

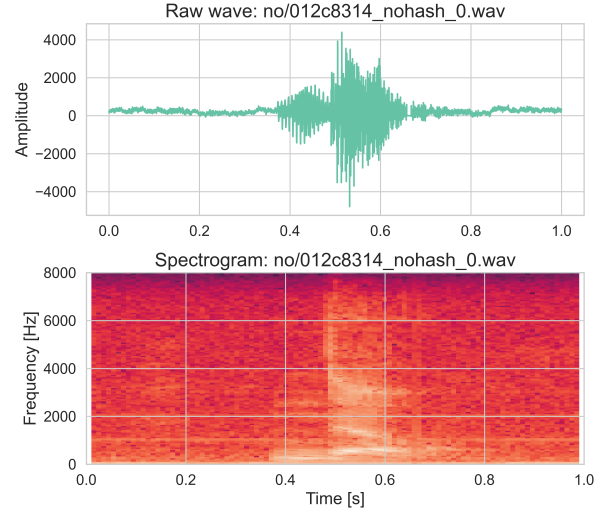


Figure 16: Raw wave and spectrogram example from train set for “no”.

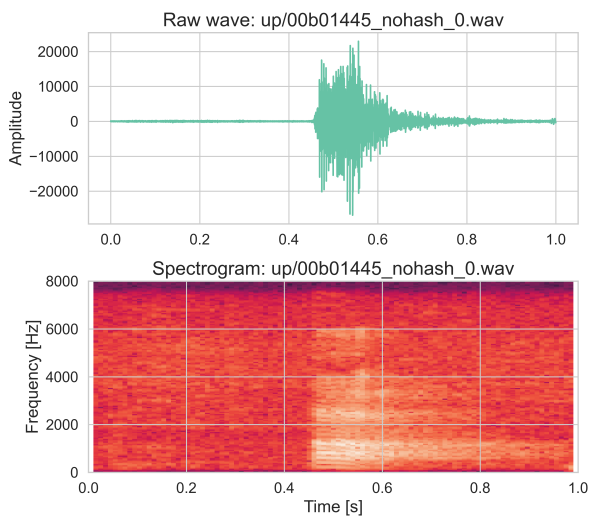


Figure 17: Raw wave and spectrogram example from train set for "up".



Figure 19: Score distribution across different strategies.

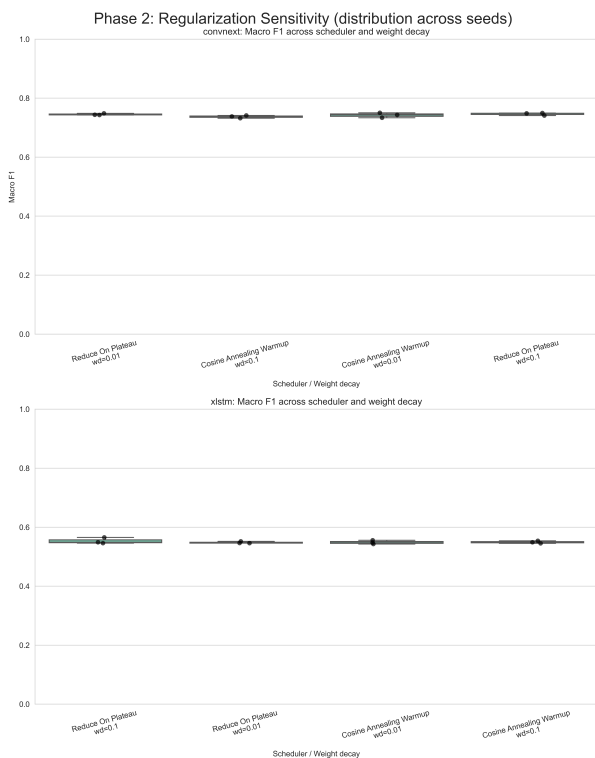


Figure 18: Regularization sensitivity across different configurations.

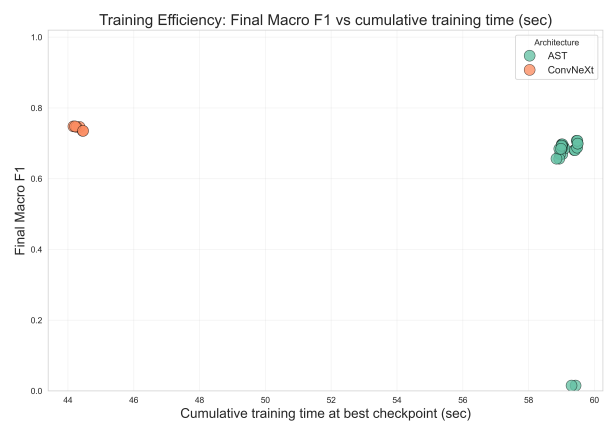


Figure 20: Training efficiency for different strategies.